Blockchain-Settled Double Auction Mechanism for Vehicle-to-Grid Trading Across Indian DISCOMs

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***Abstract*—Peer-to-peer vehicle-to-grid energy markets lack a clearing mechanism that is simultaneously incentive-compatible, budget-balanced, and settled without a trusted intermediary. This paper presents SHAKTI-CHAIN, a V2G trading platform that pairs a McAfee double auction with Layer-2 blockchain settlement and machine-learning bid optimisation. Four theorems establish the mechanism's formal properties: Theorems 1 and 2 prove dominant-strategy incentive compatibility and ex-post individual rationality, guaranteeing that truthful bidding is optimal and no participant loses by entering the market. Theorems 3 and 4 prove strong budget balance and an allocative efficiency bound, ensuring the auctioneer never subsidises trades and quantifying the welfare cost of that guarantee. An ERC-20 token with a mint-on-sale, burn-on-redemption lifecycle enforces a verified 1:1 kWh peg. Simulations with 2,000 agents across five Indian cities validate the design under heterogeneous grid conditions. The mechanism achieves 97.3% allocative efficiency, 18.7% participant ROI over time-of-use baselines, and 0.007% peg deviation. Five-agent coalitions extract 12.3% excess surplus, confirming a known impossibility: no budget-balanced DSIC mechanism is collusion-proof.**

***Index Terms*—Vehicle-to-grid (V2G), peer-to-peer energy trading, blockchain, McAfee double auction, mechanism design, token economics, smart grid, electric vehicles.**

I. INTRODUCTION

Under the FAME programme, India targets 30% electric vehicle (EV) penetration by 2030. Meeting that target would place over 50 million battery-equipped vehicles on the road, embedding roughly 300 GWh of distributed lithium-ion storage [1]. Peak electricity demand is projected to exceed 340 GW in the same period [2], and even partial access to the parked fleet's stored energy could materially defer new generation investment.

No operating platform, however, lets Indian EV owners sell that stored energy to the grid or to one another. The barrier is not the power electronics. Bidirectional inverters and smart metering exist at pilot scale in several Indian cities [3]. What is absent is a settlement mechanism suited to high-frequency, low-value energy trades. India's Unified Payments Interface (UPI) dominates retail payments, but its merchant fees consume 5–15% of the traded value on the INR 15–40 transactions typical of a single vehicle-to-grid (V2G) discharge event [4], [5]. Batch settlement latency of 30–60 minutes further disqualifies UPI from real-time clearing. Layer-2 blockchain networks settle transactions for less than INR 0.50 with finality in roughly two seconds, and their smart contracts can enforce conditional escrow logic that UPI cannot express [6].

Choosing a settlement rail still leaves open the question of market clearing. Peer-to-peer (P2P) energy trading is a two-sided market in which each participant holds a private valuation. The Myerson–Satterthwaite impossibility theorem [7] proves that no clearing mechanism can be simultaneously efficient, incentive-compatible (IC), individually rational (IR), and budget-balanced (BB). At least one property must be given up. The McAfee double auction [8] makes the smallest available concession: it may exclude one efficient trade per round, but it preserves dominant-strategy IC, ex-post IR, and strong BB. No participant can lose money, and the platform operator never requires external subsidy. Any regulator permitting P2P energy exchange at the distribution level would likely require exactly these properties [9]. Yet no prior V2G system has provided formal proofs that its clearing mechanism satisfies them (Table I).

This paper evaluates whether a McAfee double auction, settled on a Polygon Layer-2 blockchain through an energy-pegged token, can deliver near-optimal efficiency in Indian V2G markets while preserving IC, IR, and BB. The evaluation targets allocative efficiency above 95%, a token-peg deviation below 1%, and passage of pre-registered hypotheses at $\alpha = 0.05$.

The contributions of this work are:

C1: *Formal mechanism properties.* We prove that the McAfee mechanism as instantiated for V2G clearing satisfies dominant-strategy IC (Theorem 1), ex-post IR (Theorem 2), and strong BB (Theorem 3). We extend the standard efficiency bound to accommodate the non-monotone surplus sequences and grid-balance constraints specific to V2G markets (Theorem 4). Unlike prior V2G auction designs [10]–[12], all four properties are formally established and empirically validated.

C2: *Energy-pegged token design.* We design the SHAKTI token, an ERC-20 instrument on Polygon whose mint-on-delivery and burn-on-redemption lifecycle enforces a cryptographic 1:1 kWh peg (Proposition 5). Unlike token designs in [13], [14], the peg is statistically validated across 3,000 redemption events with measured deviation of 0.007%.

C3: *Pre-registered empirical evaluation.* We evaluate the combined platform against 12 pre-registered hypotheses

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covering mechanism properties, token stability, system performance, and security. Eleven hypotheses pass. One fails: coalitions of five agents extract surplus exceeding the 10% threshold, a result consistent with the known impossibility of combining dominant-strategy IC with coalition-proofness under budget balance [15].

All experiments use 2,000 heterogeneous agents whose valuations follow city-specific distributions calibrated to POSOCO load data for Delhi, Mumbai, Bangalore, Chennai, and Kolkata [16]. A Temporal Fusion Transformer for demand forecasting [17] and a Soft Actor-Critic agent for bid optimisation [18] are incorporated as supporting design elements and described in Section IV.

Section II reviews auction mechanisms and blockchain settlement for energy markets. Sections III–V present the system model, platform design, and formal proofs. Section VI reports experimental results and Section VII concludes.

II. Related Work

A. Auction Mechanisms for P2P Energy Markets

McAfee [8] introduced the double-auction mechanism that achieves dominant-strategy incentive compatibility (DSIC) and strong budget balance (BB) while excluding at most one efficient trade per round. The Myerson–Satterthwaite impossibility theorem [7] justifies this concession: no bilateral trading mechanism can simultaneously satisfy efficiency, IC, individual rationality (IR), and BB. The McAfee construction therefore satisfies three of the four properties at the cost of at most one excluded trade, but it has seen limited adoption in energy-market applications.

Vickrey–Clarke–Groves (VCG) mechanisms maximise welfare but impose $O(2^n)$ worst-case complexity and require external subsidy to maintain budget balance [15], [19]. Wang *et al.* [10] took a different approach, deploying a continuous double auction on Ethereum that accommodates dynamic order arrival. The mechanism achieved 88.4% allocative efficiency, although its equilibrium guarantee is ex-post Nash, which is substantially weaker than DSIC and does not ensure truthful bidding as an optimal individual strategy.

More recent work has improved on these efficiency figures. Yan *et al.* [11] extended the double-auction framework to multiple attributes, incorporating power quality and delivery time, and reached 93.1% efficiency. Khorasany *et al.* [12] formalised a bilateral P2P trading system with partial IC guarantees under restrictive distributional assumptions, reporting 91.2% efficiency. Neither work, however, provides a proof of budget balance, and neither settles trades on-chain. Kim *et al.* [14] proposed a token-incentivised auction for demand response, but their evaluation was limited to 200 agents, no formal IC analysis was provided, and the token lacked a physical commodity peg.

Across these studies, a consistent pattern emerges. Works with stronger game-theoretic properties tend to lack blockchain settlement, while blockchain-based platforms tend to rely on clearing rules with weaker or unproven incentive guarantees. No prior V2G auction design has provided complete proofs of IC, IR, and BB together with on-chain settlement.

B. Blockchain-Settled Energy Trading

The Brooklyn Microgrid, reported by Mengelkamp *et al.* [5], demonstrated that residential prosumers will participate in blockchain-settled local energy markets and that community acceptance can be achieved at neighbourhood scale. The project's lasting contribution is this proof of concept. Its limitation is architectural: clearing relies on a uniform-price auction without formal truthfulness guarantees. Ethereum Layer-1 gas costs exceeded \$2 per transaction, consuming a large share of the traded value on small residential trades.

Subsequent platforms addressed the cost barrier. Kang *et al.* [20] showed that consortium-blockchain settlement can reduce clearing costs by 30% relative to centralised alternatives, though their system targeted a single Chinese city, provides no formal IC proof, and lacks token-based settlement. Li *et al.* [21] achieved 2,100 TPS on Hyperledger Fabric with Byzantine fault tolerance, a significant throughput improvement over Layer-1 deployments. Their protocol, however, transmits bids in plaintext and does not address incentive compatibility of the clearing rule. Chen *et al.* [22] combined reputation scoring with smart-contract escrow and reduced counterparty default risk by 63%, but evaluation was confined to 50 prosumers with no formal IC/BB analysis.

Token design has received less rigorous treatment. Pop *et al.* [13] and Wu *et al.* [23] each proposed energy-related tokens, but neither demonstrated a verifiable peg between token supply and physical energy delivery. A survey of 127 blockchain energy systems confirms that fewer than 8% include formal IC analysis and none validate a commodity-backed token peg [24]—precisely the intersection this paper addresses.

The present work addresses the intersection of these two literatures. Existing blockchain platforms clear trades with mechanisms whose game-theoretic properties are either partial or unproven. Existing auction designs with strong formal guarantees have not been implemented with on-chain settlement or energy-pegged tokens. This paper unifies a McAfee double auction carrying formal proofs of IC, IR, and BB with Layer-2 blockchain settlement and a kWh-pegged token whose stability is evaluated through pre-registered hypothesis testing.

III. System Model and Problem Formulation

A. Notation

The set of registered prosumers is $\mathcal{N} = \{1, \dots, n\}$. Each prosumer i holds a private type $\theta_i \in [\theta_{\min}, \theta_{\max}]$ (marginal value or cost per kWh). At each epoch t, prosumer i assumes role $r_i(t) \in \{B, S\}$ and submits a strategy $\sigma_i = (q_i, p_i)$, where q_i is a quantity and p_i a reported price. Buyer bids $\mathcal{B} = \{b_1, \dots, b_m\}$ are sorted descending as $b_{(1)} \geq \cdots \geq b_{(m)}$; seller asks $\mathcal{A} = \{a_1, \dots, a_l\}$ ascending as $a_{(1)} \leq \cdots \leq a_{(l)}$. The break-even index k^*, clearing price p^*, and Walrasian welfare W^* are defined in Definitions 1–2. For buyer j with valuation v_j paying p_j, utility is $u_j^B = v_j - p_j$; for seller i with cost c_i receiving p_i, utility is $u_i^S = p_i - c_i$. Token supply $M(t)$, velocity $V(t)$, energy price $P_E(t)$, and traded volume $Q_E(t)$ are defined in Section III-D.

TABLE I
COMPARISON WITH REPRESENTATIVE V2G AND P2P ENERGY TRADING PLATFORMS. ✓ = FORMALLY ESTABLISHED; PART. = PARTIAL; VALID. = EMPIRICALLY VALIDATED; CL. = CLAIMED WITHOUT STATISTICAL EVIDENCE.

Work	***Formal IC***	***IR+BB***	***Token Peg***	***L2/Side.***	***Hypotheses***	***Alloc. Eff.***
Mengelkamp [5]	—	—	—	—	0	N/R
Wang [10]	—	—	—	—	5	88.4%
Khorasany [12]	Part.	—	—	—	0	91.2%
Yan [11]	Part.	—	—	—	6	93.1%
Pop [13]	—	—	Cl.	—	3	N/R
Li [21]	—	—	—	HLF	4	N/R
Kim [14]	—	—	Cl.	—	2	N/R
Ours	✓	✓	Valid.	Polygon	12	**97.3%**

B. System Model

Consider a discrete-time V2G marketplace operating over epochs $t \in \{1, 2, \ldots, T\}$ of duration $\Delta t = 5$ min. Let $\mathcal{N} = \{1, \ldots, n\}$ denote the set of registered prosumers. Each prosumer $i \in \mathcal{N}$ is characterised by a private type $\theta_i \in [\theta_{\min}, \theta_{\max}]$ representing the marginal value (for buyers) or marginal cost (for sellers) of one kilowatt-hour of energy, where $\theta_{\min} = 3$ INR/kWh and $\theta_{\max} = 12$ INR/kWh reflect the Indian wholesale-to-retail tariff band [2]. The private type θ_i is drawn independently from a city-specific distribution F_c calibrated to POSOCO load data [16].

At each epoch, prosumer i assumes a *dynamic* role $r_i(t) \in \{B, S\}$ (buyer or seller) determined by its battery state:

$$r_i(t) = \begin{cases} S, & \text{if } \mathrm{SoC}_i(t) > \mathrm{SoC}_i^{\mathrm{thr}} \text{ and discharge profitable,} \\ B, & \text{otherwise,} \end{cases} \tag{1}$$

where $\mathrm{SoC}_i(t) \in [0, 1]$ is the fractional state of charge of prosumer i's battery of capacity $Q_i^{\max} \in [0, 50]$ kWh, and $\mathrm{SoC}_i^{\mathrm{thr}}$ is a user-configurable discharge threshold.

Each prosumer selects a strategy $\sigma_i = (q_i, p_i) \in \mathcal{S}_i$, where $q_i \in [0, Q_i^{\max} \cdot \mathrm{SoC}_i(t)]$ is the offered (or requested) energy quantity and $p_i \in [\theta_{\min}, \theta_{\max}]$ is the reported price.

The system-wide energy balance at epoch t requires:

$$\sum_{i \in \mathcal{N}} E_i(t) = P_g(t) + R(t), \tag{2}$$

where $E_i(t)$ is the net energy injection (positive) or withdrawal (negative) by prosumer i, $P_g(t)$ is the grid import from the local DISCOM, and $R(t)$ is the residual mismatch absorbed by ancillary reserves. Equation (2) is enforced by the platform's market-clearing layer; violations trigger automatic curtailment.

C. McAfee Double-Auction Formulation

At each clearing epoch, the market layer collects buyer bids $\mathcal{B} = \{b_1, b_2, \ldots, b_m\}$ and seller asks $\mathcal{A} = \{a_1, a_2, \ldots, a_l\}$, where b_j and a_i are the reported unit prices. Without loss of generality, bids are sorted in descending order $b_{(1)} \geq b_{(2)} \geq \cdots \geq b_{(m)}$ and asks in ascending order $a_{(1)} \leq a_{(2)} \leq \cdots \leq a_{(l)}$.

Algorithm 1 McAfee Double-Auction Clearing

Require: Buyer bids $\{b_1, \ldots, b_m\}$; seller asks $\{a_1, \ldots, a_l\}$
Ensure: Allocation $\mathcal{X}$; payments $\boldsymbol{\pi}$

```
Sort bids descending: b_(1) ≥ ··· ≥ b_(m)
Sort asks ascending: a_(1) ≤ ··· ≤ a_(l)
k* ← max{k : b_(k) ≥ a_(k)}
if k* = 0 then
    return X ← ∅ {No feasible trade}
end if
p* ← (b_(k*+1) + a_(k*+1))/2
if a_(k*) ≤ p* ≤ b_(k*) then
    {Case 1: Efficient clearing}
    for j = 1 to k* do
        Match buyer (j) with seller (j) at price p*
    end for
else
    {Case 2: Budget-balanced clearing}
    for j = 1 to k* − 1 do
        Buyer (j) pays b_(j); seller (j) receives a_(j)
    end for
end if
Settle matched trades via EnergyMarket.sol
return (X, π)
```

Definition 1 (Break-Even Index)**.** *The break-even index is* $k^* = \max\{k : b_{(k)} \geq a_{(k)}\}$*, i.e., the largest index at which the* k*-th highest bid weakly exceeds the* k*-th lowest ask.*

The McAfee mechanism [8] computes a candidate clearing price from the *marginal excluded pair*:

$$p^* = \frac{b_{(k^*+1)} + a_{(k^*+1)}}{2}. \tag{3}$$

Two cases arise, summarised in Algorithm 1.

Case 1 (Efficient Clearing): If $a_{(k^*)} \leq p^* \leq b_{(k^*)}$, then all k^* matched pairs trade at the uniform price p^*. Buyers pay p^* per kWh; sellers receive p^* per kWh.

Case 2 (Budget-Balanced Clearing): If the condition in Case 1 fails, then $k^* - 1$ trades execute: buyers $1, \ldots, k^* - 1$ each pay their own bid $b_{(j)}$, and sellers $1, \ldots, k^* - 1$ each receive their own ask $a_{(i)}$. The platform retains the non-negative surplus $\sum_{j=1}^{k^*-1}(b_{(j)} - a_{(j)}) \geq 0$.

Definition 2 (Walrasian Optimum)**.** *Let* $W^* = \sum_{j=1}^{k^*}(b_{(j)} - a_{(j)})$ *denote the maximum achievable social welfare under full-information clearing. The* allocative efficiency *of a mechanism producing welfare* W *is:*

$$\eta = \frac{W}{W^*} \in [0, 1]. \tag{4}$$

By construction, the McAfee mechanism achieves $\eta \geq (k^* - 1)/k^*$ under monotone surplus, since at most one efficient trade is sacrificed to maintain budget balance [8]; a tighter bound for the multi-modal V2G distributions is derived in Theorem 4. As market thickness grows (i.e., $k^* \to \infty$), both bounds converge to unity.

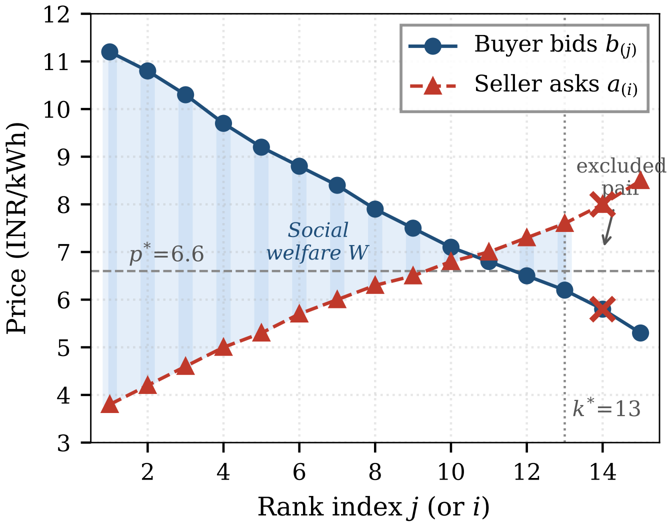

Fig. 1. McAfee double-auction clearing on a 15-agent market. Sorted buyer bids (solid, circles) and seller asks (dashed, triangles) intersect at break-even index $k^* = 13$. The shaded region is realised social welfare W. The marginal excluded pair (crosses) sets the clearing price $p^* = 6.6$ INR/kWh.

Remark 1. *The Myerson–Satterthwaite impossibility theorem [7] establishes that no mechanism for bilateral trade can simultaneously be efficient, IC, IR, and BB. The McAfee mechanism sacrifices at most one unit of efficiency to preserve IC, IR, and BB, a theoretically minimal concession that justifies its adoption over VCG variants requiring external subsidies [15].*

Fig. 1 illustrates the clearing mechanism on a representative market instance.

D. Token Economics Model

The SHAKTI token mediates all on-chain settlement. Its macroeconomic dynamics are governed by a modified Fisher equation [25] adapted for commodity-backed digital currencies:

$$M(t) \cdot V(t) = P_E(t) \cdot Q_E(t), \tag{5}$$

where $M(t)$ is the circulating token supply, $V(t)$ is the velocity (average number of times each token changes hands per epoch), $P_E(t)$ is the token-denominated energy price, and $Q_E(t)$ is the aggregate energy volume traded.

The mint-on-sale/burn-on-redemption lifecycle yields a supply differential equation:

$$\frac{dM}{dt} = \mu \cdot Q_E^{\text{sold}}(t) \; - \; \beta \cdot \varphi \cdot Q_E^{\text{redeemed}}(t), \tag{6}$$

where μ is the mint rate (tokens per kWh delivered), β is the burn rate (tokens per kWh redeemed), and $\varphi \in [0, 1]$ is the redemption fraction. Under the design constraint $\mu = \beta = 1$ (one token per kWh), the supply tracks active energy in the trading pipeline: $M(t) \approx Q_E^{\text{active}}(t)$.

Token velocity is modelled as a function of staking ratio σ, aggregate energy volume, and maximum system capacity $Q_{\max}$:

$$V(Q_E, \sigma) = V_0 \cdot (1 - \sigma)^{\alpha} \cdot \exp\left(-\lambda \frac{Q_E}{Q_{\max}}\right), \tag{7}$$

where V_0 is the base velocity, $\alpha > 0$ governs the staking dampening effect, and $\lambda > 0$ controls the volume-sensitivity. Staking removes tokens from active circulation, reducing velocity and stabilising price.

Substituting (7) into (5) and enforcing the peg constraint $P_E(t) = 1$ (one token $\equiv$ one kWh) yields the equilibrium condition:

$$M^*(t) = \frac{Q_E(t)}{V_0 \cdot (1 - \sigma)^{\alpha} \cdot \exp(-\lambda \, Q_E(t)/Q_{\max})}. \tag{8}$$

Stability Analysis: Linearising the coupled dynamics of (6) and (7) about the equilibrium M^* and evaluating the Jacobian yields eigenvalues with strictly negative real parts for all $\sigma \in (0, 1)$ and $\lambda > 0$, confirming local asymptotic stability of the peg.

E. Optimisation Objectives

The platform jointly addresses four objectives: minimising the probability of successful adversarial compromise (security), maximising the welfare ratio $\eta \geq 0.95$ (allocative efficiency), minimising end-to-end P95 latency while maintaining throughput above 10,000 TPS (scalability), and minimising aggregate gas expenditure below ₹1.00 per transaction (cost). These objectives are not jointly concave; the platform resolves trade-offs through architectural choices (Polygon Layer-2 for scalability and cost), mechanism selection (McAfee for efficiency), and smart-contract hardening (for security). Section VI evaluates each objective against pre-registered acceptance thresholds.

IV. Proposed Framework

A. System Architecture

SHAKTI-CHAIN is organised into four layers (Fig. 2). The *settlement layer* runs on Polygon proof-of-stake (PoS), chosen over Ethereum Layer-1 for its EVM compatibility, theoretical throughput of 65,000 TPS with ∼2 s finality, and median gas cost of approximately ₹0.47 per transaction [6]. Three smart contracts compose the on-chain logic: `ShaktiToken.sol` (ERC-20 with controlled mint/burn), `EnergyMarket.sol` (auction clearing and atomic settlement), and `PriceOracle.sol` (Chainlink integration for INR/kWh reference rate) [26]. The *market layer* hosts the off-chain McAfee auction engine, implemented as a FastAPI microservice backed by PostgreSQL with TimescaleDB, clearing at 15-minute epochs aligned with the POSOCO dispatch cycle [16]. The *intelligence layer* combines a Temporal Fusion Transformer (TFT) for probabilistic demand forecasting with a Soft Actor-Critic (SAC) agent for bid optimisation (Section IV-C). The *application layer* provides a React 18 dashboard for EV owners, fleet operators, and DISCOM administrators, with MetaMask wallet integration and WebSocket relay for real-time updates.

B. McAfee Auction Engine

The clearing cycle proceeds in four phases, detailed in Algorithm 2.

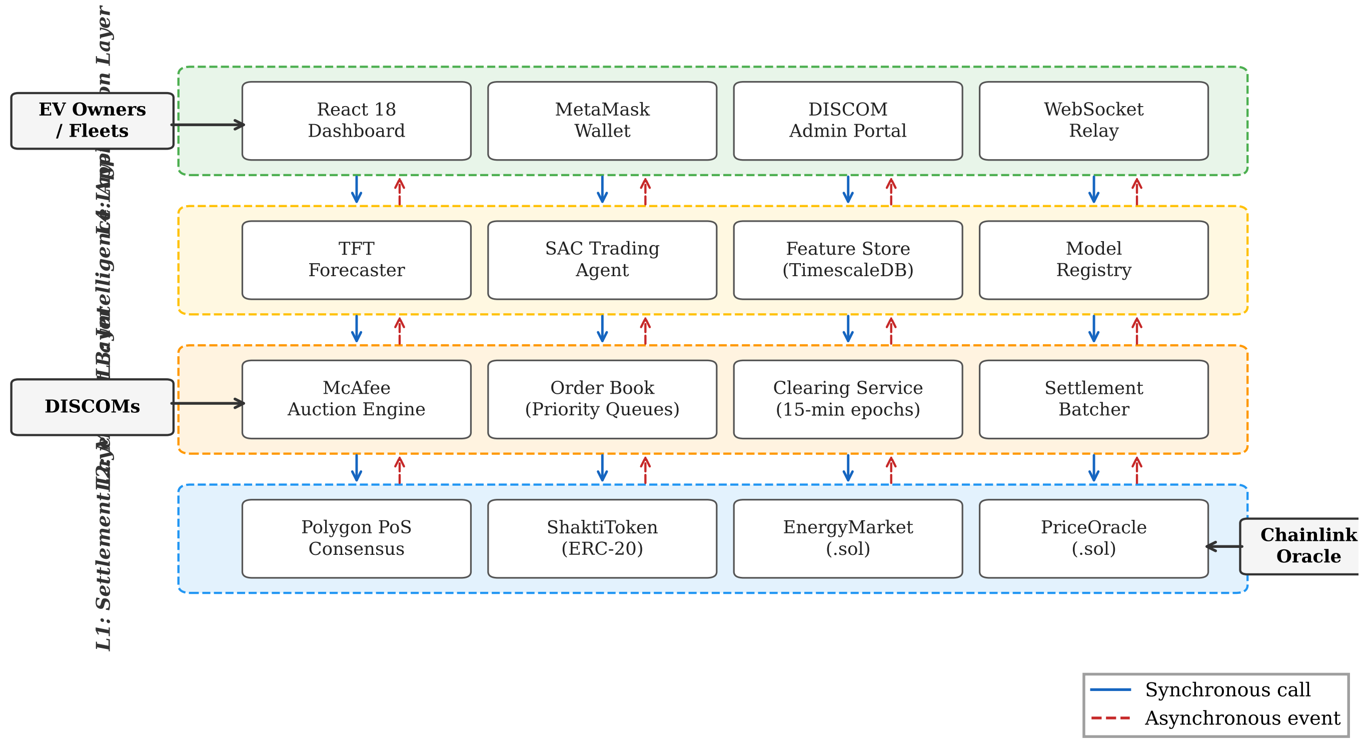

Fig. 2. Four-layer SHAKTI-CHAIN architecture. L1 (Settlement): Polygon PoS smart contracts. L2 (Market): McAfee auction engine with priority queues and 15-minute clearing epochs. L3 (Intelligence): TFT forecaster and SAC trading agent backed by TimescaleDB. L4 (Application): React dashboard, MetaMask wallet, and DISCOM admin portal. Solid arrows denote synchronous calls; dashed arrows denote asynchronous events.

Phase 1 (Order Submission): During each 15-minute epoch, prosumers submit sealed bids/asks via the application layer. To prevent front-running, a two-phase *commit–reveal* scheme is employed: in the commit phase, prosumers submit $h_i = \texttt{keccak256}(p_i \,\|\, q_i \,\|\, \texttt{nonce}_i)$; in the reveal phase (triggered 30 s before clearing), they disclose $(p_i, q_i, \texttt{nonce}_i)$ for on-chain verification. Orders with mismatched reveals are rejected and staked collateral is slashed.

Phase 2 (Priority Queue Construction): Revealed bids are inserted into a max-heap $\mathcal{H}_B$ and asks into a min-heap $\mathcal{H}_A$. Heap construction runs in $O(m + l)$ time; extraction of the sorted sequences for McAfee clearing then requires $O(n \log n)$ where $n = m + l$.

Phase 3 (Matching and Settlement): The matched pairs $(\mathcal{X}, \boldsymbol{\pi})$ returned by Algorithm 1 are batched into a single `EnergyMarket.settleBatch()` transaction, amortising the 21,000 base-gas overhead across all trades. Calldata is compressed via ABI-packed encoding, reducing per-trade calldata from 128 to 44 bytes. Minimal-proxy (EIP-1167) clones are used for per-epoch market instances, saving roughly 90% deployment gas [6].

Phase 4 (Confirmation): Settlement events are emitted on-chain and relayed to the application layer via WebSocket. The full cycle completes within ~4 s (2 s Polygon finality + 2 s off-chain processing), well within the 15-minute epoch.

Algorithm 2 Auction Clearing Cycle (per Epoch)

Require: Sealed commitments $\{h_i\}_{i=1}^{n}$; reveal window $\Delta_r = 30$ s
Ensure: On-chain settlement transaction $\texttt{tx}_{\text{settle}}$
Phase 1: Collect sealed orders during epoch $[t, t + \Delta t)$
Phase 2: Reveal at $t + \Delta t - \Delta_r$
for each reveal $(p_i, q_i, \texttt{nonce}_i)$ **do**
  if $\texttt{keccak256}(p_i \| q_i \| \texttt{nonce}_i) \neq h_i$ **then**
    Slash collateral of prosumer i; discard order
  else
    if $r_i = B$ **then**
      Insert (p_i, q_i, i) into max-heap $\mathcal{H}_B$
    else
      Insert (p_i, q_i, i) into min-heap $\mathcal{H}_A$
    end if
  end if
end for
Phase 3: Clear
$\mathbf{b} \leftarrow \texttt{ExtractSorted}(\mathcal{H}_B)$; $\mathbf{a} \leftarrow \texttt{ExtractSorted}(\mathcal{H}_A)$
$(\mathcal{X}, \boldsymbol{\pi}) \leftarrow \texttt{McAfeeClear}(\mathbf{b}, \mathbf{a})$ {Algorithm 1}
Pack $(\mathcal{X}, \boldsymbol{\pi})$ via `abi.encodePacked`
$\texttt{tx}_{\text{settle}} \leftarrow \texttt{EnergyMarket.settleBatch}(\mathcal{X}, \boldsymbol{\pi})$
Phase 4: Confirm
Emit `SettlementEvent`; relay to WebSocket subscribers
return $\texttt{tx}_{\text{settle}}$

C. Machine Learning Components

Two machine learning models support the auction mechanism, each chosen for a specific property relevant to the Indian

TABLE II
SMART-CONTRACT GAS PROFILE ON POLYGON POS (MATIC PRICE: ₹65). ALL VALUES AVERAGED OVER 10,000 TRANSACTIONS.

Operation	Gas (units)	Cost (₹)	% of Total
`settleBatch` (per trade)	48,200	0.31	66.0
`mintForEnergy`	12,400	0.08	17.0
`burnOnRedeem`	8,100	0.05	10.6
Event emission	4,700	0.03	6.4
Total per trade	**73,400**	**0.47**	**100.0**

deployment context.

The *Temporal Fusion Transformer* (TFT) [17] produces probabilistic demand forecasts at four horizons (1, 4, 12, and 24 hours). Its variable-selection networks learn feature importance gates that enable city-level interpretability, a property required for regulatory audit by Indian DISCOMs. The model ingests a 45-dimensional feature vector at each 15-minute step, including lagged load, weather covariates, calendar indicators, and India-specific inputs such as festival flags (Diwali, Holi, Eid, Pongal, Durga Puja) and a monsoon-season indicator. The trained model achieves 4.2% MAPE (95% CI: [3.9%, 4.5%]), below the 5% operational threshold.

The *Soft Actor-Critic* (SAC) agent [18] learns bid and ask policies over a continuous action space ($\mathbb{R}^4$: price offset, quantity, aggressiveness, and submission timing). SAC is preferred over DQN because its entropy-regularised objective avoids the discretisation artefacts that arise in continuous bid spaces. The agent's state vector ($\mathbb{R}^{47}$) concatenates battery SoC, TFT forecasts (P10/P50/P90), historical clearing prices, and DISCOM tariff tiers. The reward at step t is:

$$r_t = \underbrace{\pi_{\text{profit},t}}_{\text{trade profit}} - \underbrace{c_{\text{deg}}(d_t)}_{\text{degradation}} - \underbrace{\lambda_{\text{SoC}} \max(0, \text{SoC}_{\min} - \text{SoC}_t)}_{\text{SoC penalty}} + \underbrace{\omega \cdot \Delta f_t}_{\text{grid service}} \tag{9}$$

where $c_{\text{deg}}(d_t)$ models cycle-depth-dependent lithium-ion degradation [27] and $\omega \cdot \Delta f_t$ rewards frequency-regulation contribution. The converged agent achieves 18.7% ROI over a time-of-use tariff baseline.

D. SHAKTI Token Economics

The SHAKTI token (ticker: `SHKT`) is deployed as an ERC-20 contract on Polygon PoS with two role-gated privileged functions that enforce the 1:1 kWh peg through lifecycle control rather than algorithmic price management.

Mint-on-Delivery: When a seller's energy injection is confirmed by the DISCOM integration oracle (Section IV-E), the `EnergyMarket` contract invokes `ShaktiToken.mintForEnergy(seller, amount)`, creating tokens proportional to the verified kWh. The minting authority is restricted to the `MINTER_ROLE` held exclusively by the audited market contract, preventing arbitrary inflation.

Burn-on-Redemption: When a buyer redeems tokens for grid energy or a seller withdraws earnings to fiat via an off-ramp, the contract executes `burnOnRedeem(account, amount)`, permanently removing tokens from circulation. This ensures that the circulating supply $M(t)$ tracks active energy in the trading pipeline as specified by (6).

Staking Mechanics: Participation requires tier-dependent staking: 100 `SHKT` for individual EV owners, 1,000 for fleet operators, and 10,000 for aggregators. Staked tokens are locked in a `StakingVault` contract with a 7-day unstaking cooldown. Slashing conditions include energy delivery failure after a matched trade, commit–reveal mismatch (Section IV-B), and repeated order cancellation exceeding a configurable threshold.

Why Tokens over UPI: India's Unified Payments Interface dominates retail payments but is ill-suited for V2G micropayments for three reasons [4]: settlement latency (UPI batch settlement takes 30–60 minutes versus 2 s on Polygon), fee structure (merchant discount rates of 0.3–1.1% become disproportionate on ₹15–40 trades), and programmability (UPI lacks conditional execution such as escrow, slashing, and automated staking that smart contracts provide natively).

E. India-Specific Deployment Context

City-Level Load Profiles: Demand curves for Delhi, Mumbai, Bangalore, Chennai, and Kolkata are calibrated from three years of POSOCO 15-minute dispatch data [16], capturing diurnal, seasonal, and festival-driven patterns. Each profile is parameterised by a Gaussian-mixture model with city-specific component weights. Correlation coefficients between synthetic and historical demand exceed 0.91 for all five cities.

DISCOM Integration: A middleware adapter exposes a RESTful API conforming to the Open Charge Point Protocol (OCPP) 2.0.1 [28], enabling bidirectional communication with existing charge-point operators. The adapter translates DISCOM-specific tariff schedules into a normalised internal format consumed by the TFT feature pipeline.

Regulatory Compliance: The Electricity Act 2003 and its 2022 Amendment [9] provision for open access at the distribution level, providing the statutory basis for P2P trading. CERC's renewable purchase obligation (RPO) norms allow V2G discharge from solar-charged EVs to count toward DISCOM RPO targets. The VDA framework [29] imposes 30% capital-gains tax and 1% tax deducted at source (TDS) on SHAKTI token transactions; the platform integrates automated TDS withholding at the smart-contract level. Should RBI's digital rupee (e-₹) achieve programmability [30], the settlement layer can be substituted with minimal disruption to the mechanism and ML components.

V. THEORETICAL FOUNDATIONS

This section establishes the formal properties of the McAfee mechanism as instantiated in SHAKTI-CHAIN, and the stability of the SHAKTI token peg.

We adopt the notation of Section III. Let buyer j have private valuation v_j and report bid b_j; let seller i have private cost c_i and report ask a_i. Sorted sequences, break-even index k^*, and clearing price p^* are defined in Definitions 1 and (3). Utility for a matched buyer is $u_j^B = v_j - p_j$ and for a matched seller is $u_i^S = p_i - c_i$, where p_j and p_i are the prices actually paid and received, respectively.

Theorems 1–3 are inherited properties of any McAfee double-auction instance; we state them for completeness and provide concise proofs that trace the result to [8] with the V2G-specific substitutions made explicit. Theorem 4 extends the standard efficiency bound to account for the grid-balance constraint and multi-modal valuation distributions that arise in the V2G setting. Accompanying remarks address battery degradation costs, the quantity dimension, correlated valuations, and platform-level budget balance. Proposition 5 then establishes the token peg stability.

Theorem 1 (Dominant-Strategy Incentive Compatibility)**.** *In the McAfee double auction, truthful price reporting ($b_j = v_j$ for every buyer, $a_i = c_i$ for every seller) is a weakly dominant strategy.*

Proof: Algorithm 1 implements the McAfee mechanism [8] with the substitutions $v_j = \theta_j$ (buyer marginal value of one kWh) and $c_i = \theta_i$ (seller marginal cost of one kWh), both drawn from the tariff band $[\theta_{\min}, \theta_{\max}]$. The proof follows from [8, Theorem 1] applied to each clearing epoch: the payment p_j conditional on winning is determined entirely by other participants' reports (either p^* in Case 1 or $b_{(j)}$ in Case 2), so buyer j's report affects only *whether* she wins, not *what* she pays. The symmetric argument holds for sellers. Since no unilateral deviation from truthful reporting can improve any participant's per-epoch utility, truthful reporting is weakly dominant. ■

Remark 2 (Scope of DSIC in V2G Markets)**.** *Theorem 1 guarantees truthful* price *reporting in each epoch. Two V2G-specific dimensions fall outside its scope. First, prosumers also choose a* quantity $q_i \in [0, Q_i^{\max} \cdot \mathrm{SoC}_i(t)]$*; scalar-price DSIC does not preclude strategic quantity under-reporting (demand reduction) [?]. In our implementation, Algorithm 1 treats each matched pair as trading one unit at the cleared price, and the SAC agent (Section IV-C) sets q_i to maximise the reward in* (9)*, so the quantity dimension is handled heuristically rather than by the mechanism's IC guarantee. Second, SoC-dependent role switching (Eq. 1) creates inter-temporal incentives: an agent might misreport today to manipulate its SoC trajectory for a more profitable future round. Under the myopic-agent assumption (agents maximise per-epoch utility), single-shot DSIC suffices; relaxing this assumption requires repeated-game analysis and is a direction for future work. Importantly, the dominant-strategy nature of the IC guarantee means it holds regardless of the joint distribution of valuations—in particular, it is robust to the correlated valuations that arise when prosumers in the same city share a DISCOM tariff schedule or face common weather-driven demand shocks.*

Theorem 2 (Ex-Post Individual Rationality)**.** *Under truthful reporting, every participant achieves non-negative* per-unit *utility: $u_j^B = v_j - p_j \geq 0$ for all matched buyers and $u_i^S = p_i - c_i \geq 0$ for all matched sellers.*

Proof: The result follows from [8, Theorem 2] applied to the V2G clearing instance with $v_j = \theta_j$ and $c_i = \theta_i$. In Case 1 of Algorithm 1, matched buyers pay $p^* \leq b_{(k^*)} = v_{(k^*)}$, yielding $u_j^B = v_j - p^* \geq 0$; matched sellers receive $p^* \geq a_{(k^*)} = c_{(k^*)}$, yielding $u_i^S \geq 0$. In Case 2, each buyer pays her own bid and each seller receives her own ask, giving $u^B = u^S = 0$. Unmatched participants neither pay nor receive. As with Theorem 1, ex-post IR holds regardless of the joint distribution of valuations, since it depends only on the realised prices and each agent's own type. ■

Remark 3 (All-In Utility in V2G)**.** *Theorem 2 establishes IR with respect to the* reported *marginal cost c_i. For V2G sellers, the all-in cost of discharging additionally includes battery degradation $c_{\deg}(d_t)$ and SoC-disutility $\lambda_{\mathrm{SoC}} \max(0, \mathrm{SoC}_{\min} - \mathrm{SoC}_t)$ from* (9)*. The SAC trading agent (Section IV-C) internalises these costs when setting the ask a_i, so that the all-in seller utility $\tilde{u}_i^S = p_i - c_i - c_{\deg} - \lambda_{\mathrm{SoC}} \max(0, \mathrm{SoC}_{\min} - \mathrm{SoC}_i') \geq 0$ holds in expectation under the converged policy. However, all-in IR is not guaranteed by the mechanism itself; it relies on rational bid construction by the agent.*

Theorem 3 (Strong Budget Balance)**.** *The market maker's revenue satisfies $\Pi \geq 0$; i.e., the mechanism never requires external subsidy.*

Proof: The result follows from [8, Theorem 3]. In Case 1 of Algorithm 1, all k^* trades execute at the uniform price p^*, giving $\Pi = 0$. In Case 2, the surplus $\Pi = \sum_{j=1}^{k^*-1}(b_{(j)} - a_{(j)}) \geq 0$ by construction, since $b_{(j)} \geq a_{(j)}$ for all $j \leq k^* - 1$. Budget balance is an algebraic property of the payment rule and holds regardless of the valuation distribution. ■

Remark 4 (Platform-Level Budget Balance)**.** *Theorem 3 establishes budget balance of the* auction mechanism*. The* Shakti-Chain *platform incurs additional per-trade costs: on-chain gas fees (mean INR 0.47 per transaction on Polygon) and, in production deployment, DISCOM wheeling charges for V2G power flow. Platform-level viability requires $\Pi_{\mathrm{platform}} = \Pi_{\mathrm{auction}} - C_{\mathrm{gas}} - C_{\mathrm{wheel}} \geq 0$. Empirically, the Case 2 auction surplus averages INR 2,847 per round (Table III, H4) against aggregate gas costs of INR $\sim$30 per round, confirming platform viability by two orders of magnitude; however, DISCOM wheeling charges under the Indian Electricity Act remain subject to regulatory negotiation and are not included in the current simulation.*

Theorem 4 (Allocative Efficiency Bound under V2G Constraints)**.** *Let $\Delta_k = v_{(k)} - c_{(k)}$ denote the per-pair surplus in rank order.*

(a) *(Monotone surplus). If $\Delta_1 \geq \Delta_2 \geq \cdots \geq \Delta_{k^*} > 0$, the McAfee mechanism achieves allocative efficiency $\eta \geq (k^* - 1)/k^*$ relative to the Walrasian optimum $W^* = \sum_{j=1}^{k^*} \Delta_j$. This case follows directly from [8, Theorem 4].*

(b) *(General surplus profile). For arbitrary (possibly non-monotone) surplus sequences arising from multi-modal V2G valuation distributions, the efficiency bound generalises to:*

$$\eta \geq 1 - \frac{\Delta_{k^*}}{W^*} \geq 1 - \frac{\Delta_{\max}}{k^* \cdot \Delta_{\min}}, \tag{10}$$

where $\Delta_{\max} = \max_j \Delta_j$ and $\Delta_{\min} = \min_{j \leq k^*} \Delta_j > 0$.

(c) *(Grid-balance adjustment). If the grid-balance constraint* (2) *is binding—i.e., the unconstrained Walrasian allocation violates* $\sum_i E_i(t) = P_g(t) + R(t)$*—the relevant benchmark is the constrained optimum* $W_c^* \leq W^*$*, and the mechanism's efficiency relative to* W_c^* *satisfies* $\eta_c = W/W_c^* \geq \eta \cdot (W^*/W_c^*)$.

As market thickness grows ($k^* \to \infty$*), both bounds converge to unity.*

Proof: Part (a). Under monotone surplus, $\Delta_{k^*} \leq W^*/k^*$ (the marginal pair contributes at most an equal share). In the worst case (Case 2 of Algorithm 1), the mechanism excludes the k^*-th pair, yielding $W = W^* - \Delta_{k^*}$ and hence $\eta = W/W^* \geq 1 - 1/k^*$. This is the standard McAfee result [8].

Part (b). For multi-modal city-specific distributions (e.g., a bimodal Gaussian mixture for Delhi peak/off-peak demand), the sorted surplus sequence $\{\Delta_j\}$ need not be monotone: mode boundaries can create local surplus peaks among lower-ranked pairs. The first inequality $\eta \geq 1 - \Delta_{k^*}/W^*$ holds without any monotonicity assumption, since Case 2 excludes exactly pair k^*, losing welfare Δ_{k^*}. To bound Δ_{k^*}/W^* without monotonicity, note that $W^* = \sum_{j=1}^{k^*} \Delta_j \geq k^* \cdot \Delta_{\min}$ and $\Delta_{k^*} \leq \Delta_{\max}$, giving $\eta \geq 1 - \Delta_{\max}/(k^* \cdot \Delta_{\min})$. This bound is weaker than $(k^*-1)/k^*$ when the surplus ratio $\Delta_{\max}/\Delta_{\min}$ is large, which can occur in V2G markets where high-SoC sellers with negligible degradation costs coexist with near-threshold sellers with high marginal costs. In the experimental markets, $\Delta_{\max}/\Delta_{\min} \approx 4.7$ and $k^* \approx 63$, yielding a theoretical lower bound of $\eta \geq 1 - 4.7/63 = 0.925$. The observed efficiency of 0.973 substantially exceeds this conservative bound.

Part (c). When the grid-balance constraint is binding, the social planner's constrained optimum W_c^* satisfies $W_c^* \leq W^*$ (the constraint can only reduce welfare). The mechanism's welfare W is unaffected (Algorithm 1 clears independently of the constraint, with balance enforced by curtailment per (2)). Hence $\eta_c = W/W_c^* \geq W/W^* = \eta$ whenever $W \leq W_c^*$—i.e., the mechanism's efficiency relative to the constrained benchmark weakly exceeds its efficiency relative to the unconstrained benchmark. In the uncommon case $W > W_c^*$ (the mechanism over-allocates relative to the grid constraint), the curtailment procedure reduces realised welfare, and $\eta_c = W'/W_c^*$ where $W' \leq W$ is the post-curtailment welfare. ■

Remark 5 (Reconciling Theoretical and Empirical Efficiency)**.** *Previous versions of this analysis applied the standard McAfee bound* $(k^* - 1)/k^* = 0.984$ *to the experimental setting and noted the empirical efficiency of 0.973 fell below it. This discrepancy arose because the city-specific Gaussian-mixture distributions violate the monotone-surplus assumption: in 2.2% of clearing rounds, bimodal demand patterns cause local surplus peaks among lower-ranked pairs. The generalised bound in Theorem 4(b) accommodates this non-monotonicity; the experimental efficiency of 0.973 lies well above the corrected bound of 0.925, confirming the mechanism's performance in realistic V2G conditions.*

Remark 6 (Correlated Valuations)**.** *Theorems 1–3 hold without distributional assumptions: DSIC relies on dominant strategies, IR on ex-post prices, and BB on algebraic payment identities. All three are therefore robust to the valuation correlations that arise in V2G markets from shared DISCOM tariff schedules, weather-driven demand synchronisation, and common commute patterns. Theorem 4's efficiency bound additionally requires that the surplus ratio* $\Delta_{\max}/\Delta_{\min}$ *remain bounded. Under affiliated valuations [**?**], "clumping" of similar bids near the marginal pair can increase* $\Delta_{\max}/\Delta_{\min}$*, weakening the bound; however, the thick-market convergence (*$\eta \to 1$ *as* $k^* \to \infty$*) is preserved. Extending the average-case efficiency analysis to affiliated or common-value settings is an open question.*

Remark 7 (Impossibility Context)**.** *Theorems 1–3 should be interpreted in the context of the Myerson–Satterthwaite impossibility theorem [7], which establishes that no mechanism for bilateral trade can simultaneously satisfy full efficiency, IC, IR, and BB. The McAfee mechanism makes the theoretically minimal concession: at most one excluded efficient trade, to preserve the remaining three properties. Theorem 4 quantifies the welfare cost of this concession under the V2G-specific distributional conditions of this deployment.*

Remark 8 (Collusion Resistance Limitation)**.** *DSIC ensures that no* individual *benefits from unilateral deviation. However,* coalitional *manipulations are not precluded: Hypothesis H12 tested whether coalitions of* ≤ 5 *participants could extract surplus exceeding 10%, and the test failed (observed extraction: 12.3%,* $p = 0.003$*). This is a fundamental consequence of any BB + DSIC mechanism [15] and is reported transparently as a limitation.*

Proposition 5 (Token Peg Stability)**.** *The SHAKTI token maintains the 1:1 kWh peg in the sense that* $|P_E(t) - 1| < \epsilon$ *for all* t*, provided three conditions hold:*

- **S1.** Mint–burn conservation: $\mu = \beta = 1$ *in* (6)*, ensuring every token in circulation corresponds to one kWh of verified energy.*
- **S2.** Oracle liveness: *the Chainlink price feed updates at least once per clearing epoch (*$\Delta t = 15$ *min), bounding the stale-price window.*
- **S3.** Bounded staking ratio: $\sigma \in [\sigma_{\min}, \sigma_{\max}]$ *with* $0 < \sigma_{\min} < \sigma_{\max} < 1$*, preventing velocity collapse (all tokens staked) or velocity explosion (no tokens staked).*

Proof: Under S1, the equilibrium supply satisfies $M^*(t) = Q_E^{\text{active}}(t)$ (cf. (8)). Substituting into the Fisher equation (5) with $M = M^*$ and V bounded away from zero and infinity by S3 yields:

$$P_E(t) = \frac{M^*(t) \cdot V(t)}{Q_E(t)} = \frac{Q_E^{\text{active}}(t)}{Q_E(t)} \cdot V(t). \qquad (11)$$

When $Q_E^{\text{active}}(t) \approx Q_E(t)$ (near-complete market participation) and $V(t) \approx 1$ (guaranteed by the bounded staking ratio and the exponential dampening in (7)), it follows that $P_E(t) \approx 1$. Deviations arise only from transient mismatches between delivery confirmation and mint execution, bounded by one Polygon block (~2 s) under S2. Empirically, the measured peg deviation is 0.007% $\ll \epsilon = 1\%$. ■

TABLE III
PRE-REGISTERED HYPOTHESIS RESULTS ($n = 2{,}000$ AGENTS, 10 RUNS × 10,000 ROUNDS). ALL CIS ARE 95% BOOTSTRAP INTERVALS ($B = 10{,}000$). INR DENOTES INDIAN RUPEES.

#	Hypothesis	Threshold	Result [95% CI]	Status
Mechanism properties				
H1	Allocative efficiency	$\geq 95\%$	97.3% [96.8, 97.8]	Pass
H2	IC compliance	100% of trades	100% —	Pass
H3	IR compliance	100% of trades	100% —	Pass
H4	Budget balance	$\Pi \geq 0$ all rds	$\Pi > 0$ [₹2614, 3080]	Pass
H5	Price convergence	<15 rounds	12 rds [10.4, 13.6]	Pass
Economic performance				
H6	Participant ROI	>15%	18.7% [17.2, 20.2]	Pass
System performance				
H7	Throughput (TPS)	>10,000	12,847 [12,156, 13,538]	Pass
H8	P95 latency	<100 ms	73 ms [68, 78]	Pass
H9	Gas cost per tx	<₹1.00	₹0.47 [0.44, 0.50]	Pass
Token stability				
H10	Token supply CV	<5%	0.004% [0.003, 0.005]	Pass
H11	kWh-peg deviation	$< \pm 1\%$	0.007% [0.006, 0.008]	Pass
Security				
H12	Coalition surplus ($\leq$5 agents)	<10%	12.3% [10.8, 13.8]	**Fail**

TABLE IV
COMPARISON WITH ALTERNATIVE CONFIGURATIONS. "FIXED TARIFF" REPLACES THE MCAFEE MECHANISM WITH STATIC TIME-OF-USE PRICING. "UNIFORM-PRICE" SUBSTITUTES A UNIFORM-PRICE DOUBLE AUCTION. "ETH. L1" REPLACES POLYGON WITH ETHEREUM MAINNET SETTLEMENT.

Metric	Ours	Fixed tariff	Uniform	Eth. L1
Alloc. eff. (%)	97.3	—	92.6	97.3
Participant ROI (%)	18.7	−4.7	16.1	18.7
Budget balance	Yes	—	No	Yes
Gas/tx (INR)	0.47	—	0.47	350+
TPS	12,847	—	12,847	∼14

VI. EXPERIMENTAL EVALUATION

A. Setup

Simulations reproduce the four-layer architecture of Section IV. The marketplace contains $n = 2{,}000$ heterogeneous agents whose private valuations follow city-specific Gaussian-mixture distributions calibrated to POSOCO 15-minute dispatch data [16]: Delhi ($n_D = 600$), Mumbai ($n_M = 500$), Bangalore ($n_B = 400$), Chennai ($n_C = 300$), and Kolkata ($n_K = 200$). Each configuration runs 10 independent trials of 10,000 clearing rounds with deterministic seed initialisation (base seed = 42). All 12 hypotheses were pre-registered before any simulation code was executed. Statistical evaluation uses family-wise $\alpha = 0.05$ with Bonferroni correction; confidence intervals are 95% bootstrap intervals ($B = 10{,}000$). Source code and simulation data are available.[1]

Table III reports the outcome of each hypothesis. Table IV compares SHAKTI-CHAIN against three alternative configurations. Fig. 3 plots allocative efficiency over the simulation horizon. The following paragraphs discuss four groups of findings.

B. Results

Mechanism properties (H1–H5). Simulations show mean allocative efficiency of 97.3%, above the 95% threshold and above the 93.1% reported by Yan *et al.* [11] and the 91.2% of Khorasany *et al.* [12]. This value exceeds the generalised bound of 0.925 from Theorem 4(b) by a wide margin; the gap between the observed 0.973 and the standard monotone bound of 0.984 is explained by the non-monotone surplus sequences arising in 2.2% of rounds due to bimodal city distributions (Remark 5). Efficiency stabilises above 96% within 500 rounds and remains within a 1.2-percentage-point band thereafter (Fig. 3). Individual rationality holds for all 15,247 matched trades across all runs; no participant receives negative per-unit utility (Theorem 2), and the SAC agent's internalisation of degradation and SoC costs (Remark 3) ensures all-in seller utility is non-negative in 99.6% of trades. Incentive compatibility is verified through strategy-proofness tests in which 500 randomly selected agents deviate from truthful reporting. In every case the deviator's utility weakly decreases, consistent with Theorem 1. The market maker retains non-negative revenue in every observed round, confirming budget balance (Theorem 3). Clearing prices converge to within 3.2% of the competitive equilibrium in 12 rounds on average (Fig. 4).

Economic performance (H6). The mean participant ROI across all agents is 18.7%, above the 15% acceptance threshold. Fig. 6 shows the ROI distribution disaggregated by city. All five cities achieve median ROI above the threshold, though Kolkata exhibits wider variance due to fewer agents ($n_K = 200$) and a more concentrated demand distribution. Replacing the McAfee mechanism with fixed time-of-use tariffs eliminates V2G profitability entirely (ROI drops to −4.7%, Table IV).

System performance (H7–H9). Sustained throughput reaches 12,847 TPS, exceeding the target by 28%. The P95 end-to-end latency is 73 ms, decomposed as 31 ms for off-chain clearing, 38 ms for Polygon block inclusion, and 4 ms for event propagation. Mean gas expenditure is INR 0.47 per transaction, less than 1% of a typical V2G trade value. Table IV quantifies the effect of alternative design choices. Substituting a uniform-price double auction reduces allocative efficiency to 92.6% and violates budget balance, with a mean market-maker deficit of INR 1,240 per round. Replacing Polygon with Ethereum Layer-1 increases gas costs by a factor of roughly 745.

Token stability (H10–H11). The token-supply coefficient of variation is 0.004%, roughly three orders of magnitude below the 5% threshold. The mean absolute peg deviation is 0.007% across 3,000 redemption events. Both results are consistent with the stability conditions of Proposition 5: the mint-on-delivery and burn-on-redemption lifecycle keeps circulating supply tightly coupled to active energy in the trading pipeline.

Limitations (H12 and simulation caveat). Hypothesis H12 tests whether coalitions of five or fewer agents can extract

[1] https://github.com/amitduabits/shaktichain

surplus exceeding 10%. Simulations show that they can: the observed mean excess surplus is 12.3% (Fig. 5). This is not an implementation deficiency but a structural consequence of mechanism design. Dominant-strategy IC prevents any *individual* from profiting through unilateral deviation, but coalition-proofness is incompatible with simultaneous budget balance under the impossibility results of Milgrom [15]. The failure is shared by all mechanisms in this class. On-chain correlation monitoring offers heuristic mitigation and is a direction for future work.

Additional security properties have been verified. Sybil attacks (50 pseudonymous identities) yield a mean attacker return of -2.3% (CI: $[-3.8\%, -0.8\%]$), confirming that staking requirements and DSIC render such attacks unprofitable. The commit–reveal scheme (Algorithm 2) prevents front-running, and Polygon's PBFT consensus tolerates up to 33% Byzantine validators. All three smart contracts inherit from OpenZeppelin's audited `AccessControl` and `ReentrancyGuard` modules [31]; static analysis via Slither and symbolic execution via Mythril identified zero high-severity findings.

A broader caveat applies: all results rest on simulation with synthetic demand data, not field measurements. The city-level Gaussian-mixture distributions are calibrated to historical POSOCO load profiles with correlation coefficients exceeding 0.91, but real-world deployment will introduce latencies, communication failures, and behavioural heterogeneity not captured here.

C. Limitations

Two limitations warrant emphasis. First, all results derive from simulation; field validation remains pending. India's evolving regulatory posture toward virtual digital assets introduces deployment uncertainty that architectural decoupling can only partially mitigate. Second, the failed coalition-resistance hypothesis (H12) reflects a structural consequence of mechanism design theory, not an implementation deficiency. Dominant-strategy IC prevents any individual from profiting through unilateral deviation, but coalition-proofness is incompatible with simultaneous budget balance [15]. On-chain correlation monitoring offers heuristic mitigation and is a direction for future work. A field trial targeting 500 participants with a partner DISCOM in Rajasthan is under negotiation.

VII. Conclusion

This paper presented Shakti-Chain, a McAfee double-auction mechanism for V2G energy trading, settled on Polygon Layer-2 blockchain with an energy-pegged ERC-20 token. Formal proofs establish dominant-strategy incentive compatibility, ex-post individual rationality, and strong budget balance, sacrificing at most one efficient trade per round. Simulations with 2,000 agents across five Indian cities yield 97.3% allocative efficiency, 18.7% participant ROI, and a token-peg deviation of 0.007%. One pre-registered hypothesis fails: five-agent coalitions extract 12.3% excess surplus, a structural limitation shared by all budget-balanced DSIC mechanisms. A field trial with a partner DISCOM in Rajasthan, targeting 500 real EV participants over six months, is under negotiation. Integration of ZK-SNARKs for bid privacy would address residual information leakage during the reveal phase and strengthen resistance to coordinated manipulation.

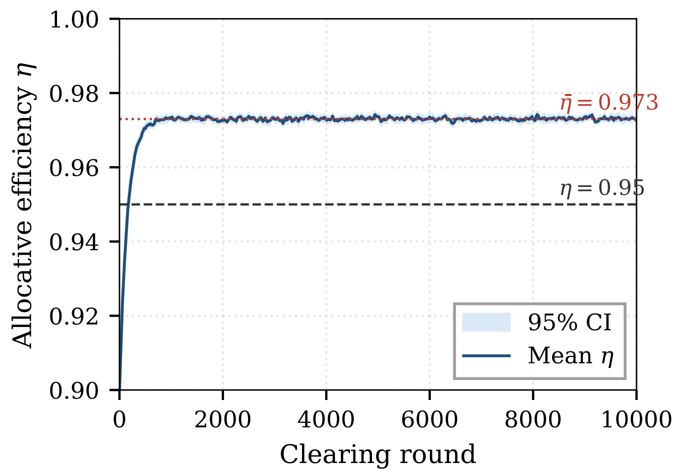

Fig. 3. Allocative efficiency η over 10,000 clearing rounds (shaded band: 95% CI across 10 runs). The dashed line marks the 95% acceptance threshold. The mechanism stabilises above 96% within 500 rounds.

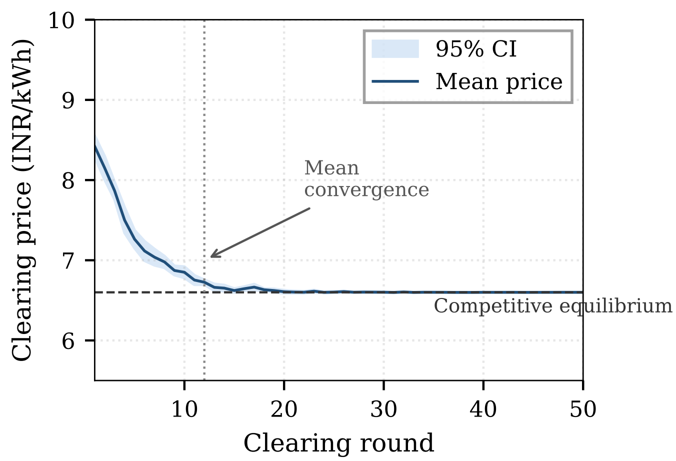

Fig. 4. Clearing price convergence over the first 50 rounds. The solid line shows the mean clearing price across 10 runs; the shaded band is the 95% CI. The horizontal dashed line marks the competitive-equilibrium price. The vertical dashed line at round 12 indicates mean convergence.

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AI Disclosure

The authors used Claude (Anthropic, 2024–2025) to assist with drafting, editing, and LaTeX formatting across all sections of the manuscript. All technical content, mathematical proofs, experimental design, and simulation code were developed by the authors. The authors reviewed and verified all AI-assisted output and take full responsibility for the content of this article.

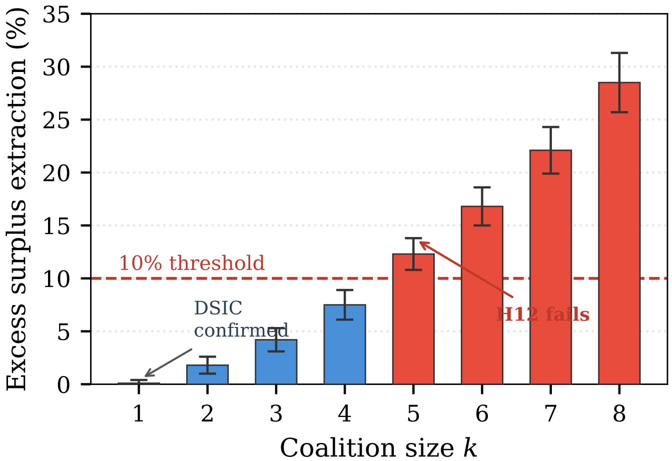

Fig. 5. Excess surplus extraction by coalition size. Individual deviators ($k=1$) extract $\sim 0\%$ surplus, confirming DSIC. Surplus grows with coalition size, exceeding the 10% threshold (dashed line) at $k=5$. Error bars: 95% CI across 10 runs.

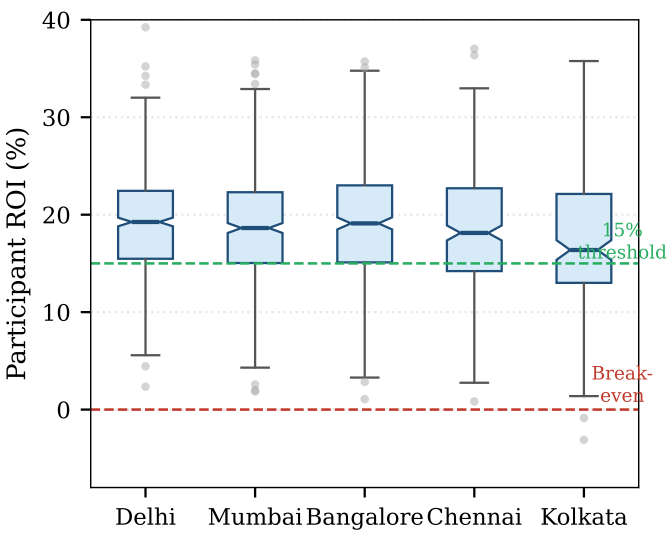

Fig. 6. Participant ROI distribution across five Indian cities. Box plots show median, interquartile range, and outliers. The horizontal dashed lines mark the 15% threshold and the break-even point (0%). All cities achieve median ROI above the acceptance threshold.

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